


Figure 21: Refusal tuning performance across three types of samples (Mistral-7B-v0.3).

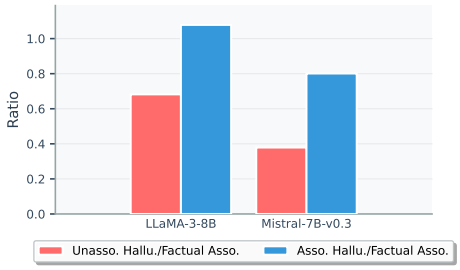



Figure 22: Comparison of subspace overlap ratios.

### D.2 Quantitative Analysis of Cluster Separability

To quantitatively validate the visual separability observed in the t-SNE visualizations (Figures 7 and 14), we compute two cluster separability metrics on the hidden representations: the Silhouette coefficient and the Davies-Bouldin Index (DBI). The Silhouette coefficient ranges from $-1$ to 1, where higher values indicate better-defined clusters. The DBI measures the average similarity ratio of each cluster with its most similar cluster, where lower values indicate better separation.

The results shown in Table 4 quantitatively confirm our observations from the t-SNE visualizations. When comparing UHs and FAs, both models exhibit high Silhouette coefficients (0.33–0.40) and low DBI values (0.93–1.13), indicating clear cluster separation between UHs and FAs. In contrast, when comparing AHs and FAs, the Silhouette coefficients are near zero (0.0022 and $-0.0011$) and DBI values are substantially higher (14.22 and 25.25), indicating strong overlap between AHs and FAs. These metrics support our claim that FAs and UHs form well-separated clusters in the representation space, while AHs strongly overlap with FAs, consistent with our hypothesis that AHs rely on similar knowledge-recall processes as FAs.

| | UH Only | | AH Only | |
|---|---|---|---|---|
| **Model** | **Sil. ↑** | **DBI ↓** | **Sil. ↑** | **DBI ↓** |
| LLaMA | 0.397 | 0.932 | 0.002 | 14.22 |
| Mistral | 0.327 | 1.134 | -0.001 | 25.25 |

Table 4: Quantitative cluster separability metrics. Higher Silhouette (Sil., ↑) and lower Davies–Bouldin Index (DBI, ↓) indicate better separation.

| **Model** | **Method** | **UH Only** | **AH Only** |
|---|---|---|---|
| Mistral | Prob. | .894 (.889–.899) | .457 (.453–.461) |
| | Subj. Pop. | .812 (.787–.839) | .570 (.551–.590) |
| | Subject | .893 (.878–.909) | .592 (.552–.631) |
| | Attention | .872 (.864–.881) | .584 (.500–.668) |
| | Last Token | .923 (.913–.934) | .629 (.610–.648) |
| LLaMA | Prob. | .860 (.853–.867) | .488 (.480–.495) |
| | Subj. Pop. | .869 (.856–.883) | .479 (.467–.491) |
| | Subject | .913 (.897–.930) | .648 (.626–.669) |
| | Attention | .920 (.889–.950) | .579 (.525–.634) |
| | Last Token | .935 (.923–.947) | .691 (.660–.722) |

Table 5: Hallucination detection performance with 95% confidence intervals across five random seeds.

### D.3 More Visualization on Hidden States

In this section, we provide t-SNE visualization of subject tokens' hidden states in Figure 18 and Figure 19.

Compared to the last-token representations, the t-SNE visualization of subject-token hidden states shows that unassociated hallucinations (UHs) are moderately separated from factual and associated samples, they exhibit marginally lower separability than last-token representations. This observation aligns with the results in §5, where the hallucination detection performance using last-token hidden states outperforms that based on subject-token representations.

## E Further Analysis on Hallucination Detection

### E.1 Statistical Significance Analysis

We report the hallucination detection performance (§5) with 95% confidence intervals across five random seeds in Table 5. Across all settings relevant to our claims, AUROC on UHs is significantly higher than on AHs (all $p < 0.01$), supporting our finding that current detection methods are substantially more effective for UHs than AH.

### E.2 Semantic Consistency-Based Detection

We sample 200 instances per category and generate 5 responses per prompt with temperature = 0.7. We

| Model | UH Only | AH Only |
|---|---|---|
| LLaMA | 0.8374 | 0.6183 |
| Mistral | 0.8437 | 0.6159 |

Table 6: Semantic consistency-based hallucination detection performance.

| Threshold | UH Only | | AH Only | |
|---|---|---|---|---|
| | Sensitivity | Specificity | Sensitivity | Specificity |
| 0.1 | 0.900 | 0.833 | 0.697 | 0.583 |
| 0.3 | 0.879 | 0.888 | 0.644 | 0.633 |
| 0.5 | 0.867 | 0.925 | 0.601 | 0.679 |
| 0.7 | 0.867 | 0.942 | 0.534 | 0.725 |
| 0.9 | 0.838 | 0.963 | 0.433 | 0.771 |

Table 7: Sensitivity-specificity trade-offs at different thresholds.

compute a consistency score for each sample and evaluate hallucination detection performance using these scores, measured by AUROC. Table 6 shows the results.

The results show that the performance of this consistency-based method aligns with our findings for both the black-box and white-box methods discussed in §5: it can effectively distinguish UHs from FAs (AUROC $\approx 0.84$) but struggles to separate AHs from FAs (AUROC $\approx 0.62$). This behavioral evidence supports our mechanistic analysis: since AHs rely on similar knowledge-recall processes as FAs, they tend to produce similarly consistent outputs across runs, whereas UHs exhibit higher variance due to weaker reliance on subject representations.

### E.3 Sensitivity-Specificity Trade-offs

To further examine the detection performance beyond AUROC, we analyze the trade-offs between sensitivity and specificity at various classification thresholds. We report results for detectors trained on LLaMA's last-token representations. Tables 7 present the results.

For UH Only, we observe that detection performance is remarkably robust across the threshold range. Even as the threshold increases from 0.1 to 0.9, the detector maintains a high balance of metrics, with sensitivity remaining above 0.83 and specificity exceeding 0.96 at the highest setting. This stability indicates a clear and consistent separation in the representation space. In contrast, for AH Only, the performance is highly sensitive to the specific threshold. As the threshold moves from 0.1 to 0.9, sensitivity drops significantly from 0.697 to 0.433, while specificity only improves from 0.583 to 0.771. This trade-off suggests that establishing a stable decision boundary between AHs and FAs is challenging, confirming that their internal representations strongly overlap.